

RS004 Week 12 — English Worksheet

Topic: Coin Types + Spin Animation (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

This week coins come in **kinds** — gold (1 point), gem (5), secret (10). Each coin is stored as a **dictionary** with its rect, kind, and value, and they **spin** with a `sin` wave that wobbles their width. On this worksheet you think about and explain that code in English — you don't type or run anything.

Keep these words handy: **dictionary**, **kind / value**, `math.sin`, **wobble (animate)**, **frame**.

1 · Predict 🧠

Read each snippet and write what you think it does.

```
coins = [
    {"rect": pygame.Rect(180, 480, 20, 20), "kind": "gold", "value": 1},
    {"rect": pygame.Rect(580, 480, 20, 20), "kind": "gem", "value": 5},
]
```

Each coin is now a dictionary. How do you read the second coin's value?


```
wave = math.sin(frame * 0.1) * 8
w = max(2, int(20 + wave))
```

`math.sin(...)` swings between -1 and +1. What is the smallest the width `w` can get, and why the `max(2, ...)` ?


```
if player.colliderect(coin["rect"]):  
    score = score + coin["value"]
```

The player grabs a gem. How much does the score go up?

2 · Read the Dictionary

A coin is `{"rect": pygame.Rect(450, 250, 20, 20), "kind": "secret", "value": 10}`. Fill in the blanks.

```
coin["kind"]    = ____  
coin["value"]   = ____  
coin["rect"].x = ____
```


Why is a dictionary better than three separate lists (one for rects, one for kinds, one for values)?

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — Every coin should add its own value, but all coins add 1 point no matter the kind.

```
for coin in coins[:]:
    if player.colliderect(coin["rect"]):
        score = score + 1
        coins.remove(coin)
```

Hint: use `coin["value"]`, not a fixed `1`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The collision check should compare the player with the coin's rectangle, but it crashes saying a dict can't collide.

```
for coin in coins[:]:
    if player.colliderect(coin):
        score += coin["value"]
```

Hint: the rect lives inside the dictionary under a key.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The coins should keep wobbling, but they spin once and then freeze.

```
def draw_coin(coin, frame):  
    wave = math.sin(frame * 0.1) * 8  
    w = max(2, int(20 + wave))  
    ...  
# in the game loop:  
frame = 0  
draw_coin(coin, frame)
```

Hint: `frame` never increases, so `sin` never changes. Where should `frame` grow?

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Read this working coin-types code, then answer the questions.

```
coins = [  
    {"rect": pygame.Rect(180, 480, 20, 20), "kind": "gold", "value": 1},  
    {"rect": pygame.Rect(580, 480, 20, 20), "kind": "gem", "value": 5},  
    {"rect": pygame.Rect(450, 250, 20, 20), "kind": "secret", "value": 10},  
]  
  
def draw_coin(coin, frame):  
    wave = math.sin(frame * 0.1) * 8  
    w = max(2, int(20 + wave))  
    rect = coin["rect"]  
    pygame.draw.ellipse(screen, (255, 215, 0), (rect.x, rect.y, w, rect.height))  
  
# in the game loop:  
for coin in coins[:]:  
    draw_coin(coin, frame)  
    if player.colliderect(coin["rect"]):  
        score = score + coin["value"]  
        coins.remove(coin)  
frame = frame + 1
```

Why is each coin written as a dictionary instead of just a rect?

What does `coin["value"]` add to the score for a gem, and for a secret coin?

How does `math.sin(frame * 0.1)` make the coin look like it is spinning?

Why is `max(2, ...)` used when setting the width `w` ?

Why does `frame = frame + 1` need to run every loop?

5 · Explain Your Lesson Code 🎥

Explain the coin code **you wrote in today's lesson**. Record a short video on your phone. You may show it running. Try to use these words: **dictionary**, **kind**, **value**, **sin**, **spin**.

1. Show your gold, gem, and secret coins.
2. Collect each one and show the score jump by different amounts.
3. Read your `math.sin` line out loud and explain the wobble.
4. Show the secret coin and the jump to reach it.

Write what you will say in your video. Plan it here before you record.

Submit ✓

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.